WORKED SOLUTION

STEP II 1992, Question 12

Mechanics • Hooke's Law

STEP II 1992, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q12

The key idea

Each wheel can turn freely, so the only condition for it to be in equilibrium is that the moments about its centre balance: the hanging particle tries to turn the wheel one way and the elastic string pulls it back the other way. The geometry gives the extension of the elastic string in terms of θ, so Hooke's law gives the tension, and taking moments gives one equation in θ. Counting equilibria then means counting solutions of that equation for the angles that actually make sense.

Solution

Setting up the model

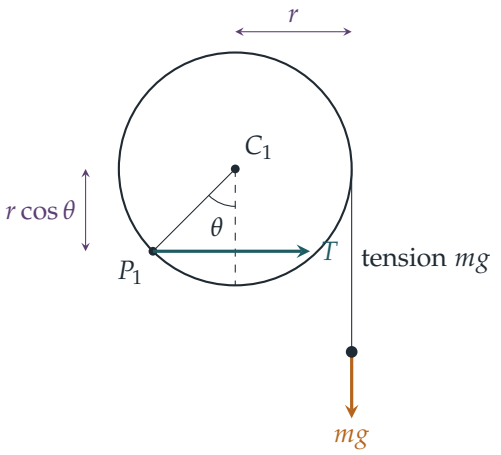
We model the particles as particles, the strings carrying them as light and inextensible, and the elastic string as light and obeying Hooke's law. The axles are smooth, so the reaction at each centre has no moment about that centre.

Take C_1C_2 as horizontal. In the figure, P_1 lies a distance $r\sin\theta$ outside C_1 (to the left) and $r\cos\theta$ below it; P_2 is the mirror image. So P_1 and P_2 are at the same height and the elastic string is horizontal, with length

$$P_1P_2 = d + 2r\sin\theta.$$

Its natural length is d, so the extension is $2r\sin\theta$ and, by Hooke's law, the tension is

$$T = \frac{\lambda\,(2r\sin\theta)}{d}.$$



The particle on W_1 hangs in equilibrium, so its string has tension mg; that string leaves the wheel tangentially on the side nearer W_2, at horizontal distance r from C_1. The elastic string pulls P_1 horizontally towards W_2 with force T, and its line of action is a distance $r\cos\theta$ below C_1.

The equilibrium condition

Taking moments about C_1: the weight turns W_1 clockwise (as drawn) with moment mgr, and the tension turns it anticlockwise with moment $Tr\cos\theta$. So

$$mgr = Tr\cos\theta = \frac{2\lambda r\sin\theta}{d}\, r\cos\theta = \frac{\lambda r^2 \sin 2\theta}{d},$$

using $2\sin\theta\cos\theta = \sin 2\theta$. Dividing by λr^2 gives

$$\sin 2\theta = \frac{mgd}{\lambda r},$$

as required. By symmetry the same equation comes from taking moments about C_2.

Which angles are allowed?

> **Ask yourself** Why can't θ be any angle at all, so that $\sin 2\theta = k$ has infinitely many solutions?

The moment of the tension must oppose the weight. If $\sin\theta \le 0$, the points P_1 and P_2 are no more than d apart, so the elastic string is slack, $T = 0$, and nothing balances mgr. If $\cos\theta \le 0$, the string runs level with or above C_1, so its moment about C_1 is zero or turns the wheel the same way as the weight. Either way there is no equilibrium. Hence an equilibrium needs

$$0 < \theta < \tfrac{\pi}{2}, \qquad \text{that is,} \qquad 0 < 2\theta < \pi.$$

(Could the two wheels sit at *different* angles? The moment of the tension about each centre is T times the perpendicular distance from that centre to the line P_1P_2, and both moments must equal mgr. So C_1 and C_2 are the same distance from the line and on the same side of it, below the centres. Hence the line is parallel to C_1C_2, so P_1 and P_2 are at the same depth and the two angles agree, because if they had opposite signs the string would be exactly length d and slack. So every equilibrium is of the symmetric kind shown.)

Write $k = \dfrac{mgd}{\lambda r} > 0$. On $0 < 2\theta < \pi$, $\sin 2\theta$ rises from 0 to 1 at $2\theta = \frac{\pi}{2}$ and falls back to 0, taking every value in $(0, 1)$ exactly twice and the value 1 exactly once. So the equation $\sin 2\theta = k$ has

- no solution if $k > 1$;
- exactly one solution, $\theta = \frac{\pi}{4}$, if $k = 1$;
- exactly two solutions, $\theta = \alpha$ and $\theta = \frac{\pi}{2} - \alpha$ with $\alpha = \frac{1}{2}\arcsin k \in (0, \frac{\pi}{4})$, if $0 < k < 1$.

Since $k > 1 \iff \lambda < \dfrac{mgd}{r}$, the answers are:

Part (i)

No equilibrium positions when $\lambda < \dfrac{mgd}{r}$: the elastic string is too weak to hold the particles at any angle.

Part (ii)

Just one equilibrium position when $\lambda = \dfrac{mgd}{r}$, with $\theta = \frac{\pi}{4}$.

Part (iii)

Exactly two equilibrium positions when $\lambda > \dfrac{mgd}{r}$, at $\theta = \alpha$ and $\theta = \frac{\pi}{2} - \alpha$.

Part (iv)

There is no value of λ that gives more than two equilibrium positions, because θ is confined to $(0, \frac{\pi}{2})$.

Checks

- Dimensions: λ is a force and so is mg, and d/r is a ratio of lengths, so $mgd/(\lambda r)$ is dimensionless, as a sine must be.
- Limiting case: as $\lambda \to \infty$ the string becomes inextensible, and the two solutions tend to $\theta \to 0$ (tiny extension, large tension) and $\theta \to \frac{\pi}{2}$ (string almost level with the

centres, so a tiny moment arm). Both make sense.

- Numerically, with $m = 1$, $d = 4$, $r = 1$, $g = 9.8$ and $\lambda = 3mgd/r$, so $k = \frac{1}{3}$, a full model of both wheels (allowing the two angles to differ) has exactly two equilibria, at $\theta \approx 0.1699$ and $\theta \approx 1.4009$; these are $\frac{1}{2}\arcsin\frac{1}{3}$ and $\frac{\pi}{2}$ minus it.

Common slips

- Taking the extension as $r\sin\theta$ rather than $2r\sin\theta$: both ends of the string move out.
- Using the moment arm $r\sin\theta$ for the tension. The tension is horizontal, so its arm is the *vertical* distance $r\cos\theta$.
- Answering (iv) with "infinitely many" by forgetting that only $0 < \theta < \frac{\pi}{2}$ is physically possible.

Beyond STEP

Counting equilibria as a parameter changes is the start of *bifurcation theory*. Here the two equilibria for $\lambda > mgd/r$ merge at $\lambda = mgd/r$ and disappear, which is called a saddle-node (or fold) bifurcation. Looking at the potential energy, one of the pair turns out to be stable and the other unstable, which is typical of a fold. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers moments, rigid-body equilibrium and stability from potential energy.

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